PRISM: Evaluating Privacy-Constrained Customer Churn Classification

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Abstract

Predicting customer churn is vital for subscription services, but standard pipelines rely on continuous, invasive tracking of user activity. Under strict privacy regulations and cold-start conditions, collecting granular behavioral telemetry is often infeasible. We present **PRISM** (Privacy-Restricted Information for Subscriber Modeling) to investigate churn prediction under constrained feature visibility. We measure how performance degrades as behavioral features are removed, testing whether basic, non-intrusive account metadata can reliably identify at-risk subscribers. Repository: https://github.com/ibac03/PRISM-churn-privacy.

1 Introduction

Customer retention is vital for subscription business models, as churn directly reduces recurring revenue. While early churn detection enables proactive retention offers, standard pipelines assume unrestricted access to fine-grained user telemetry, including app logs and interaction histories. However, strict privacy regulations (e.g., GDPR, CCPA) and user tracking opt-outs increasingly limit the behavioral data companies can store. This project investigates whether data mining models can accurately detect churn when restricted to non-intrusive billing and administrative attributes.

2 Motivation and Significance

Retaining existing customers is significantly more cost-effective than acquiring new ones [1]. Detecting churn without invasive tracking provides two key benefits: (1) **Privacy Compliance & Trust:** Minimizing data collection mitigates regulatory penalties and preserves consumer trust [2]; and (2) **Cold-Start Feasibility:** New subscribers lack interaction history, rendering telemetry-heavy models ineffective during early contract phases.

2.1 Motivating Example

Consider a subscriber who opts out of app tracking. Because the provider possesses no activity logs, telemetry-dependent classifiers cannot evaluate this user. If a model can reliably predict churn using only administrative terms (e.g., month-to-month contract, electronic check payment, and tenure), the provider can intervene with targeted retention offers without invading user privacy.

2.2 Gaps in Existing Solutions

Most churn studies engineer dense features from usage logs to maximize benchmark scores [3], overlooking settings where telemetry is restricted. Consequently, there is limited understanding of the performance trade-offs of using a privacy-preserving subsets of data. PRISM addresses this gap by formalizing and benchmarking churn classification across restricted visibility tiers.

3 Formal Problem Definition

Following the course taxonomy, customer churn prediction is formulated as a **Boolean Classification** task under a **Fully Supervised Learning** framework.

Let O denote the set of customer observations, and let $C = \{0, 1\}$ represent the target classes, where $c = 1$ (Positive Class) indicates churn and $c = 0$ (Negative Class) indicates retention. The task is to learn a mapping function:

$$f : O \to C \quad \text{such that} \quad f(o) = c, \quad \forall o \in O,\ c \in C \tag{1}$$

where each $o \in O$ represents a customer described across restricted attribute spaces (from basic billing records to full service configurations).

3.1 Running Example Instance

Consider an observation $o \in O$ representing a customer account: ⟨Tenure: 2 mos, Contract: Monthly, Payment: E-check⟩. The learned function maps this observation to the positive class:

$$f(o) = 1 \quad \text{(Customer Churn)}$$

This flags an at-risk subscriber based strictly on administrative settings, requiring no private usage telemetry.

4 Team and Collaboration Justification

Our team comprises three undergraduate students (A. Majumdar, D. Parsana, R. Vashi) and one graduate student acting as project lead (B. Irukulapati). We formed this group around a shared interest in privacy-aware machine learning. Our collective skills across software development, data pipelines, and machine learning complement each other, enabling balanced collaboration on implementation, experimentation, and milestone deliverables.

References

[1] S. Gupta et al. 2006. Modeling customer lifetime value. *Journal of Service Research* 9, 2 (2006), 139–155.
[2] K. Martin and N. Murphy. 2017. The role of data privacy in marketing. *J. Acad. Mark. Sci.* 45, 2 (2017), 135–155.
[3] W. Verbeke et al. 2012. New insights into churn prediction in the telecommunication sector. *EJOR* 218, 1 (2012), 211–229.

A AI-Assisted Editing Disclosure (Category C)

In accordance with course guidelines for Category C (AI-Assisted Editing), generative AI was used solely to assist with LaTeX formatting, syntax adjustments, and layout condensing to satisfy the strict single-page limit. All problem formulations, conceptual ideas, team details, and definitions were developed by the team members.

A.1 Original Unedited Student Draft

Please refer to the original draft document submitted along with this document.